Week -3

Assignment Questions



PW SKILLS

Assignment Questions

1.Write a Python program to print the Fibonacci series up to n terms.

2.Write a program that takes the input of three sides of Triangle ,and print whether it is equilateral, isosceles,Scalene.

3.Write a program to print and calculate the sums of even and odd integers of the first n natural Numbers.

Example:

Input: Up to which natural number? 25

Output:

The sum of even integers is 156

The sum of odd integers is 169

4.Write the code for printing this pattern:

a)

```
1
11
111
1111
11111
```

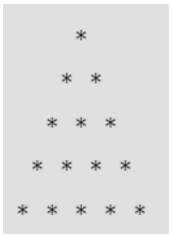
b.

```
4321
432
43
4
```

C.

```
1
1 3
1 3 5
1 3 5 7
```

D.



E.

```
*********
 *******
  *****
   ***
    *
```

5.Write a program to separate Even, Odd, Prime, Composite Numbers from a given List, make a separate list for each one.

6.Write a program to calculate and print the distance traveled by an object moving in uniform circular motion.
Hint: distance=2×π×radius×number of revolutions

7.Write a short program to input a digit and print it in words.
8.Write a short program to check whether the square root of a number is prime or not.

9.Use a suitable data structure for storing Player name and Jersey number of the players of Indian Women's Cricket team. and Display it on the basis of their Jersey Number.
Hint: use a dictionary.

Here are the jersey numbers of Indian women cricket team players.

Player	Jersey number
Rajeshwari Gayakwad	1
Anjum Chopra, Sneh Rana	2
Mithali Raj	3
Jemimah Rodrigues	5
Reema Malhotra, Deepti Sharma	6
Harmanpreet Kaur	84, 7
Ekta Bisht	8
Renuka Singh Thakur	10
Yastika Bhatia	11
Shikha Pandey	12
Richa Ghosh	13
Punam Raut	14
Shafali Verma	17
Smriti Mandhana	18
Radha Yadav	21
Pooja Vastrakar	23
Poonam Yadav	24
Jhulan Goswami	25
Meghna Singh	26
Sabbineni Meghana	27
Taniya Bhatia	28
Mona Meshram	30
Priya Punia	69
Veda Krishnamurthy	79
Harleen Deol	98

10.Write a program to take N (N > 20) as an input from the user. Print numbers from 11 to N. When the number is a multiple of 3, print "Chahal", when it is a multiple of 7, print "Thala". When it is a multiple of both, print "ChahalThala".

11. Write programs using nested loops
(a)
A
A B
A B C D
A B C D E
A B C D E F

(b)
0
2 2
4 4 4
8 8 8 8

12. Write a program using nested loops to produce a rectangle of "s with 6 rows and 20 *s per row.

13.Given a DNA sequence represented as a string, write a program using a for loop to count the occurrence of each nucleotide (A, T, C, and G) in the sequence. Print out the count of each nucleotide.
Example:
Input_sequence:ATCGATCGATCGTACGCTAGCTAGCTAGCTAGCTAGCTAGCTAGCTAGCTA
Output:
A: 13
T: 13
C: 13
G: 12

14. Given three numbers A, B and C, write a program to write their values in an ascending order. For example, if A=12, B=10, and C=15,
your program should print out:
Smallest number = 10
Next higher number = 12 Highest number = 15
Note: Don't use any python inbuilt function, Use only Loop.

15 . Write a Python script to input temperature. Then ask them what units, Celsius or Fahrenheit, the temperature is in. Your program should convert the temperature to the other unit. The conversions are F=(9/5)*C+32 and C = 5/9*(F-32).

16 . Ask the user to enter a temperature in Celsius. The program should print a message based on the temperature:

- If the temperature is less than -273.15, print that the temperature is invalid because it is below absolute zero.
- If it is exactly - 273.15, print that the temperature is absolute 0.
- If the temperature is between -273.15 and 0, print that the temperature is below freezing.
- If it is 0, print that the temperature is at the freezing point.
- If it is between 0 and 100, print that the temperature is in the normal range.
- If it is 100, print that the temperature is at the boiling point.
- If it is above 100, print that the temperature is above the boiling point.